

Engineering Physics Lab

Engineering Physics Lab is a course that is usually taken by engineering students in their first year of study. The course is designed to provide students with a practical understanding of the fundamental principles of physics that are relevant to engineering.

Some of the key topics that are covered in an Engineering Physics Lab course include:

1. **Mechanics:** This includes the study of motion, forces, and energy. Students will learn about concepts such as Newton's laws of motion, work, and energy, and will perform experiments to verify these principles.
2. **Electricity and Magnetism:** This includes the study of electric charges, electric fields, and magnetic fields. Students will learn about concepts such as Coulomb's law, Ohm's law, and Faraday's law, and will perform experiments to verify these principles.
3. **Waves and Optics:** This includes the study of wave phenomena such as sound and light. Students will learn about concepts such as wave interference, diffraction, and polarization, and will perform experiments to verify these principles.
4. **Thermodynamics:** This includes the study of heat, temperature, and the laws of thermodynamics. Students will learn about concepts such as heat transfer, entropy, and the Carnot cycle, and will perform experiments to verify these principles.

Overall, the Engineering Physics Lab course provides students with a hands-on approach to learning the fundamental principles of physics that are essential for a successful career in engineering. Students will develop their experimental skills and will learn how to analyze and interpret data, which are valuable skills for any engineer.

List of Experiments

1. **The Milgram Experiment:** This experiment was conducted by Stanley Milgram in 1961 to study obedience to authority figures. The experiment involved participants administering electric shocks to a "learner" when they answered questions incorrectly.
2. **The Stanford Prison Experiment:** This experiment was conducted by Philip Zimbardo in 1971 to study the psychological effects of becoming a prisoner or prison guard. The experiment involved college students being randomly assigned to play the role of either a prisoner or a guard in a simulated prison environment.
3. **The Asch Conformity Experiment:** This experiment was conducted by Solomon Asch in 1951 to study the effects of group pressure on individual behavior. The experiment involved participants being asked to match

the length of a line with comparison lines, while in a group setting where the other group members gave obviously incorrect answers.

4. **The Pavlov's Dog Experiment:** This experiment was conducted by Ivan Pavlov in the 1890s to study classical conditioning. The experiment involved training a dog to associate the sound of a bell with being fed, resulting in the dog salivating at the sound of the bell even when no food was present.
5. **The Bobo Doll Experiment:** This experiment was conducted by Albert Bandura in 1961 to study the effects of observing aggressive behavior. The experiment involved children observing an adult model behaving aggressively towards a Bobo doll, and then being given the opportunity to interact with the doll themselves.

Any ten experiments (at least four from each group)

Here are ten experiments, with at least four from each group:

Group 1: Physics

1. **The Double-Slit Experiment:** This experiment demonstrates the wave-particle duality of light and matter.
2. **The Michelson-Morley Experiment:** This experiment was designed to detect the presence of the hypothetical luminiferous aether, a medium thought to be necessary for the propagation of light.
3. **The Millikan Oil Drop Experiment:** This experiment measured the charge of the electron.
4. **The Rutherford Gold Foil Experiment:** This experiment demonstrated the existence of the atomic nucleus.

Group 2: Chemistry

1. **The Iodine Clock Reaction:** This experiment demonstrates the concepts of reaction rates and chemical kinetics.
2. **The Briggs-Rauscher Oscillating Reaction:** This experiment demonstrates the concepts of oscillating reactions and chemical kinetics.
3. **The Belousov-Zhabotinsky Reaction:** This experiment demonstrates the concepts of oscillating reactions and chemical kinetics.
4. **The Thermite Reaction:** This experiment demonstrates the concepts of exothermic reactions and redox reactions.

Group 3: Biology

1. **The Hershey-Chase Experiment:** This experiment demonstrated that DNA, not protein, is the genetic material of life.
2. **The Meselson-Stahl Experiment:** This experiment demonstrated that DNA replication is semi-conservative.

Group A

Group A is a term that can refer to many different things depending on the context. Here are some possible meanings of Group A:

1. In a classification system, Group A may refer to the first or primary group of items, individuals, or categories.
2. In sports, Group A may refer to a specific group of teams or athletes in a tournament or competition.
3. In medicine, Group A may refer to a specific group of diseases or conditions, such as Group A streptococcus.
4. In chemistry, Group A may refer to a specific group of elements in the periodic table.
5. In business, Group A may refer to a specific group of companies or products.

1. To determine the wavelength of sodium light by Newton's ring experiment.

- Newton's ring experiment is a method to determine the wavelength of light using the phenomenon of interference.
- In this experiment, a plano-convex lens is placed on a flat glass plate, and a monochromatic light source, such as sodium light, is used to illuminate the setup.
- The light waves reflected from the top and bottom surfaces of the air film between the lens and the glass plate interfere with each other, producing a pattern of bright and dark rings known as Newton's rings.
- The radius of the rings can be measured using a traveling microscope, and the wavelength of the light can be calculated using the formula:
$$r^2 = \frac{(D_n^2 - D_m^2)}{4R(n-m)}$$
 where r is the wavelength of the light, D_n and D_m are the diameters of the n th and m th dark rings, R is the radius of curvature of the lens, and n and m are integers.
- This experiment is named after Sir Isaac Newton, who first described the phenomenon of interference in his book "Opticks" in 1704.

2. To determine the wavelength of different spectral lines of mercury light using plane

- Spectral lines are produced when an element is heated and its electrons are excited to higher energy levels.
- When the electrons return to their ground state, they emit light at specific wavelengths, producing spectral lines.
- The wavelengths of these spectral lines are characteristic of the element and can be used to identify it.
- In this experiment, a mercury lamp is used as the light source.
- The light from the mercury lamp is passed through a diffraction grating, which separates the light into its component wavelengths.

- The diffraction grating produces a series of bright lines, known as the spectrum, on a screen.
- The position of each line in the spectrum is measured and used to calculate the wavelength of the corresponding spectral line.
- This experiment can be used to determine the wavelengths of the different spectral lines of mercury light.

Transmission Grating

- Transmission gratings are designed to disperse incident light at specific angles.
- Angular dispersion is a function of angle of incidence and groove spacing.
- Dispersion increases as angle of incidence increases or as groove spacing decreases .
- Transmission gratings are manufactured in the same way as ruled reflection gratings in which a replica of a master grating is created by duplicating its grooved surface with a thin, vacuum deposition-coated layer, except they have no reflective overcoat .
- Transmission gratings offer low polarization sensitivity as opposed to reflection gratings because incident light is not reflected by a mirror coating such as aluminum .
- Since light is transmitted through the grating, transmission gratings can be used in compact, in-line configurations .
- Transmission gratings can be volume Bragg gratings or surface relief gratings .
- Transmission gratings operate to angularly disperse incident light into a spectrum .
- The product line of transmission gratings by Coherent utilizes state-of-the-art design and precise lithographic wafer-scale fabrication to deliver record-breaking optical performance .

3. To determine the specific rotation of cane sugar solution using polarimeter.

A polarimeter is an instrument used to measure the angle of rotation of plane-polarized light as it passes through a sample of a substance. The specific rotation of a substance is defined as the observed angle of rotation, α , divided by the path length, l , in decimeters, and the concentration, c , in grams per milliliter.

The specific rotation of a substance can be calculated using the formula:

$$[\alpha]_T = \alpha / (l * c)$$

Where: - $[\alpha]_T$ is the specific rotation of the substance at a given wavelength, λ , and temperature, T . - α is the observed angle of rotation. - l is the path length in decimeters. - c is the concentration in grams per milliliter.

To determine the specific rotation of a cane sugar solution using a polarimeter, the following steps can be followed:

1. Prepare a solution of cane sugar with a known concentration.
2. Fill a polarimeter tube with the cane sugar solution and place it in the polarimeter.
3. Adjust the polarimeter to the desired wavelength and temperature.
4. Measure the angle of rotation of the plane-polarized light as it passes through the cane sugar solution.
5. Use the formula above to calculate the specific rotation of the cane sugar solution.

It is important to note that the specific rotation of a substance can vary with changes in wavelength and temperature. Therefore, it is important to specify the wavelength and temperature at which the specific rotation was determined.

4. To determine the focal length of the combination of two lenses separated by a distance

1. The focal length of the combination of two lenses separated by a distance can be determined using the lens formula.
2. The lens formula is given by $1/f = 1/v - 1/u$, where f is the focal length of the lens, v is the distance of the image from the lens, and u is the distance of the object from the lens.
3. To determine the focal length of the combination of two lenses, the distance between the lenses must be known.
4. The distance between the lenses can be measured using a ruler or a vernier caliper.
5. Once the distance between the lenses is known, the focal length of the combination of the two lenses can be calculated using the formula $1/f = 1/f_1 + 1/f_2 - d/f_1f_2$, where f_1 and f_2 are the focal lengths of the two lenses, and d is the distance between the lenses.
6. The focal length of the combination of the two lenses can also be determined experimentally by measuring the distance of the image formed by the combination of the two lenses from the lenses, and using the lens formula to calculate the focal length of the combination of the two lenses.
7. It is important to note that the focal length of the combination of two lenses is not simply the sum of the focal lengths of the two lenses. The distance between the lenses must be taken into account when calculating the focal length of the combination of the two lenses.

Verify the formula for the focal length of combination of lenses

When two thin lenses are placed in contact, the combined focal length f of the lenses can be calculated using the formula:

$$1/f = 1/f_1 + 1/f_2$$

where f_1 and f_2 are the focal lengths of the individual lenses.

This formula can be derived using the lens maker's formula and the principle of image formation by lenses. The lens maker's formula states that the power P of a lens is given by:

$$P = 1/f = (n-1)(1/R_1 - 1/R_2)$$

where n is the refractive index of the lens material, R_1 and R_2 are the radii of curvature of the two surfaces of the lens, and f is the focal length of the lens.

When two lenses are placed in contact, the image formed by the first lens becomes the object for the second lens. The distance between the object and the first lens is equal to the focal length of the first lens, f_1 . The distance between the image formed by the first lens and the second lens is equal to the focal length of the second lens, f_2 .

Using the principle of image formation by lenses, the combined focal length f of the two lenses can be calculated as:

$$1/f = 1/f_1 + 1/f_2$$

This formula can be extended to the case of more than two lenses placed in contact. The combined focal length of n lenses placed in contact is given by:

$$1/f = 1/f_1 + 1/f_2 + \dots + 1/f_n$$

where $f_1, f_2, \dots, f_n$ are the focal lengths of the individual lenses.

This formula can be used to calculate the combined focal length of any number of lenses placed in contact. It is a useful tool for designing optical systems and for understanding the behavior of lenses in combination.

5. To measure attenuation in an optical fiber

Attenuation in an optical fiber refers to the reduction in the intensity of light as it travels through the fiber. This reduction in intensity is due to absorption, scattering, and other loss mechanisms. Measuring attenuation in an optical fiber is important to ensure that the signal transmitted through the fiber is of sufficient strength to be detected at the receiving end.

Here are the steps to measure attenuation in an optical fiber:

1. **Set up the equipment:** To measure attenuation, you will need a light source, an optical power meter, and connectors to attach the fiber to the light source and power meter.
2. **Calibrate the equipment:** Before taking any measurements, it is important to calibrate the equipment to ensure accurate results. This involves setting the reference power level on the power meter to the output power of the light source.

3. **Measure the reference power:** Connect the fiber to the light source and power meter, and measure the reference power level. This is the power level that will be used as a baseline for comparison.
4. **Introduce attenuation:** Introduce attenuation into the fiber by adding a known amount of loss, such as a splice or a connector. Alternatively, you can bend the fiber to introduce loss.
5. **Measure the attenuated power:** Measure the power level after the attenuation has been introduced.
6. **Calculate the attenuation:** The attenuation is calculated by subtracting the attenuated power level from the reference power level and expressing the result in decibels (dB).

By following these steps, you can accurately measure the attenuation in an optical fiber. It is important to regularly measure attenuation to ensure that the fiber is functioning properly and that the signal transmitted through the fiber is of sufficient strength.

6. To determine the wavelength of He-Ne laser light using single slit diffraction.

1. The experiment is performed using a He-Ne laser, a single slit, and a screen.
2. The He-Ne laser emits a beam of light that passes through the single slit.
3. The light diffracts as it passes through the slit and forms an interference pattern on the screen.
4. The distance between the central maximum and the first minimum of the interference pattern is measured.
5. The wavelength of the laser light can be calculated using the formula: $\lambda = ax / L$, where λ is the wavelength, a is the width of the slit, x is the distance between the central maximum and the first minimum, and L is the distance between the slit and the screen.
6. The experiment can be repeated with different slit widths to obtain more accurate results.
7. The calculated wavelength can be compared to the known wavelength of He-Ne laser light to verify the accuracy of the experiment.

7. To study the polarization of light using He-Ne laser light.

Polarization refers to the orientation of the oscillations of an electromagnetic wave. Light is an electromagnetic wave, and its oscillations can be oriented in different directions. When light is polarized, its oscillations are oriented in a specific direction.

A He-Ne laser is a type of gas laser that uses a mixture of helium and neon gases as the lasing medium. The light emitted by a He-Ne laser is highly coherent and monochromatic, making it ideal for studying the polarization of light.

To study the polarization of light using a He-Ne laser, the following steps can be followed:

1. Set up the He-Ne laser and turn it on, allowing it to warm up for a few minutes.
2. Place a polarizer in the path of the laser beam. A polarizer is an optical device that transmits light of a specific polarization and blocks light of other polarizations.
3. Rotate the polarizer and observe the changes in the intensity of the laser beam. The intensity of the laser beam will vary as the polarizer is rotated, depending on the orientation of the polarizer relative to the polarization of the laser light.
4. Record the intensity of the laser beam as a function of the angle of the polarizer.
5. Analyze the data to determine the polarization of the laser light.

By following these steps, one can study the polarization of light using a He-Ne laser. This experiment can provide valuable insights into the properties of polarized light and its interactions with optical devices.

8. To determine the wavelength of sodium light with the help of Fresnel's bi-prism.

Fresnel's bi-prism is an optical instrument used to determine the wavelength of light. It consists of two prisms placed base to base, forming a bi-prism. When a beam of light is passed through the bi-prism, it is split into two beams, which interfere with each other to produce an interference pattern.

To determine the wavelength of sodium light using Fresnel's bi-prism, the following steps can be followed:

1. Set up the bi-prism experiment by placing the bi-prism, a sodium lamp, and a screen in a straight line.
2. Adjust the position of the bi-prism and the screen to obtain a clear interference pattern on the screen.
3. Measure the distance between the bi-prism and the screen, as well as the distance between the central bright fringe and a fringe of known order.
4. Use the formula for the fringe width to calculate the wavelength of the sodium light.

The formula for the fringe width is given by:

$$w = \frac{D}{d}$$

where w is the fringe width, λ is the wavelength of the light, D is the distance between the bi-prism and the screen, and d is the distance between the virtual sources.

By measuring the fringe width and the distance between the bi-prism and the screen, the wavelength of the sodium light can be calculated. This method is a

simple and accurate way to determine the wavelength of light.

9. To determine the coefficient of viscosity of a given liquid.

Viscosity is a measure of a fluid's resistance to flow. The coefficient of viscosity is a property of a fluid that determines the amount of internal resistance to flow. There are several methods to determine the coefficient of viscosity of a given liquid, including:

1. **Ostwald viscometer:** This method involves measuring the time it takes for a fixed volume of liquid to flow through a narrow tube under the influence of gravity. The coefficient of viscosity can be calculated using the Hagen-Poiseuille equation.
2. **Falling sphere viscometer:** This method involves measuring the terminal velocity of a sphere falling through a liquid. The coefficient of viscosity can be calculated using Stokes' law.
3. **Rotational viscometer:** This method involves measuring the torque required to rotate a spindle at a constant speed in a liquid. The coefficient of viscosity can be calculated using the relationship between torque, spindle geometry, and rotational speed.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific properties of the liquid being tested and the accuracy required for the measurement. It is important to carefully calibrate the equipment and follow standard procedures to obtain accurate results.

10. To determine the value of acceleration due to gravity (g) using compound pendulum.

1. The compound pendulum is a rigid body swinging in a vertical plane about a horizontal axis passing through the body.
2. The time period of a compound pendulum is given by the formula $T = 2\pi\sqrt{I/mgd}$, where I is the moment of inertia of the pendulum about the axis of rotation, m is the mass of the pendulum, g is the acceleration due to gravity, and d is the distance between the center of mass of the pendulum and the axis of rotation.
3. To determine the value of g using a compound pendulum, the time period of the pendulum is measured for different values of d .
4. A graph is plotted between T^2 and d , and the slope of the graph is used to calculate the value of g .
5. The value of g can be calculated using the formula $g = 4\pi^2 I / (m \cdot \text{slope})$.
6. The experiment should be repeated for different values of d to obtain an accurate value of g .
7. The value of g obtained from the experiment can be compared with the standard value of g to check the accuracy of the experiment.

8. Precautions should be taken while performing the experiment, such as ensuring that the pendulum swings in a vertical plane and that the amplitude of oscillation is small.
9. The experiment should be performed in a controlled environment to minimize the effect of external factors such as air resistance.
10. The compound pendulum method is a simple and accurate method to determine the value of g . It is widely used in laboratories and educational institutions.

Group B

Group B is a term that can refer to many different things depending on the context. Here are some possible interpretations of the term “Group B”:

1. In the context of rally racing, Group B was a set of regulations introduced by the Fédération Internationale de l'Automobile (FIA) in 1982 for competition vehicles in rallying and sports car racing. Group B cars were known for their power, speed, and lack of restrictions, which led to many accidents and fatalities. The class was eventually banned in 1986.
2. In the context of microbiology, Group B Streptococcus (GBS) is a type of bacteria that can cause illness in people of all ages. In newborns, GBS can cause sepsis, pneumonia, and meningitis. In adults, it can cause urinary tract infections, skin infections, and other illnesses.
3. In the context of vitamins, Group B refers to the B vitamins, which are a group of eight water-soluble vitamins that play important roles in cell metabolism. These vitamins include thiamine (B1), riboflavin (B2), niacin (B3), pantothenic acid (B5), pyridoxine (B6), biotin (B7), folic acid (B9), and cobalamin (B12).
4. In the context of sports tournaments, Group B may refer to one of the groups in the group stage of a tournament. Teams in Group B would compete against each other to determine which teams advance to the next stage of the tournament.

1. To determine the energy band gap of a given semiconductor material.

- The energy band gap of a semiconductor material is the difference in energy between the top of the valence band and the bottom of the conduction band.
- The band gap is a crucial property of a semiconductor material as it determines its electrical conductivity and optical properties.
- The band gap can be determined experimentally using various methods such as absorption spectroscopy, photoluminescence spectroscopy, and electrical measurements.

- In absorption spectroscopy, the absorption coefficient of the material is measured as a function of the photon energy. The onset of absorption corresponds to the band gap energy.
- In photoluminescence spectroscopy, the material is excited with light of a known energy and the emitted light is analyzed to determine the band gap.
- Electrical measurements such as temperature-dependent conductivity or Hall effect measurements can also be used to determine the band gap of a semiconductor material.
- The band gap of a semiconductor material can also be calculated theoretically using methods such as density functional theory or tight-binding calculations.

2. To study Hall effect and determine Hall coefficient, carrier density and mobility of a given semiconductor material

The Hall effect is the production of a voltage difference across an electrical conductor, transverse to an electric current in the conductor and to an applied magnetic field perpendicular to the current. It was discovered by Edwin Hall in 1879.

The Hall effect can be used to determine the Hall coefficient, carrier density, and mobility of a given semiconductor material. These properties are important for understanding the electrical behavior of the material.

To determine these properties, the following steps are typically followed:

1. A sample of the semiconductor material is prepared in the form of a thin rectangular plate.
2. The sample is placed in a magnetic field perpendicular to the plate.
3. A current is passed through the sample in a direction perpendicular to both the magnetic field and the plate.
4. The voltage difference between the two sides of the plate is measured.
5. The Hall coefficient, carrier density, and mobility can be calculated from the measured voltage difference, current, magnetic field strength, and sample dimensions.

The Hall coefficient is a measure of the strength of the Hall effect in a material. It is defined as the ratio of the induced electric field to the product of the current density and the magnetic field strength.

The carrier density is the number of charge carriers (electrons or holes) per unit volume in the material. It can be calculated from the Hall coefficient and the elementary charge.

The mobility is a measure of how easily the charge carriers can move through the material. It can be calculated from the Hall coefficient, the carrier density, and the conductivity of the material.

In summary, the Hall effect can be used to determine important electrical properties of semiconductor materials, including the Hall coefficient, carrier density, and mobility. These properties are important for understanding the behavior of the material in electronic devices.

Given Semiconductor Material Using Hall Effect Setup

The Hall effect is a phenomenon that occurs in a conductor when an electric current flows through it in the presence of a magnetic field. The magnetic field causes the electrons in the conductor to experience a force, which results in a voltage difference across the conductor. This voltage difference is known as the Hall voltage.

The Hall effect can be used to determine several properties of a given semiconductor material, including:

1. **Carrier concentration:** The carrier concentration of a semiconductor material refers to the number of free electrons or holes available for conduction. The Hall voltage is directly proportional to the carrier concentration, so by measuring the Hall voltage, the carrier concentration can be determined.
2. **Type of semiconductor:** The sign of the Hall voltage can be used to determine whether a given semiconductor material is a p-type or n-type semiconductor. A positive Hall voltage indicates that the material is a p-type semiconductor, while a negative Hall voltage indicates that it is an n-type semiconductor.
3. **Mobility:** The mobility of a semiconductor material refers to how easily the charge carriers can move through the material. The Hall voltage is inversely proportional to the mobility, so by measuring the Hall voltage, the mobility of the material can be determined.

To determine these properties using the Hall effect, a Hall effect setup is used. This typically consists of a sample of the semiconductor material, a magnetic field, and a means of measuring the Hall voltage. The sample is placed in the magnetic field and a current is passed through it. The Hall voltage is then measured and used to determine the properties of the material.

In summary, the Hall effect can be used to determine several important properties of a given semiconductor material using a Hall effect setup. These properties include the carrier concentration, type of semiconductor, and mobility of the material. By measuring the Hall voltage, these properties can be determined and used to better understand the material.

3. To determine the variation of magnetic field with the distance along the axis of a current

The magnetic field produced by a current-carrying wire can be determined using the Biot-Savart law. This law states that the magnetic field at a point is directly proportional to the current and inversely proportional to the square of the distance from the wire.

To determine the variation of the magnetic field with distance along the axis of a current-carrying wire, the following steps can be followed:

1. Set up the apparatus, including a current-carrying wire, a magnetic field sensor, and a power supply.
2. Turn on the power supply and adjust the current to the desired value.
3. Place the magnetic field sensor at different distances along the axis of the wire and record the magnetic field readings.
4. Plot a graph of the magnetic field against the distance from the wire.
5. From the graph, it can be observed that the magnetic field decreases as the distance from the wire increases.

This experiment can be used to verify the inverse square law relationship between the magnetic field and the distance from the wire. It can also be used to determine the magnetic field strength at different points along the axis of the wire.

Carrying Coil and Estimate the Radius of the Coil

A carrying coil is a coil of wire that carries an electric current. The magnetic field produced by the current in the coil can be used to estimate the radius of the coil.

Here are the steps to estimate the radius of the coil:

1. Measure the magnetic field strength at the center of the coil using a Gauss-meter.
2. Determine the number of turns in the coil.
3. Use the formula for the magnetic field at the center of a coil to calculate the radius of the coil. The formula is $B = (\mu_0 * I * N) / (2 * R)$, where B is the magnetic field strength, μ_0 is the permeability of free space, I is the current, N is the number of turns, and R is the radius of the coil.
4. Solve the equation for R to find the radius of the coil.

This method can be used to estimate the radius of a carrying coil. It is important to note that the accuracy of the estimate depends on the accuracy of the measurements and the assumptions made in the calculation.

4. To verify Stefan's law by electric method.

Stefan's law states that the total energy radiated per unit surface area of a black body across all wavelengths per unit time is directly proportional to the fourth power of the black body's thermodynamic temperature.

To verify Stefan's law by electric method, the following steps can be followed:

1. Set up the apparatus, which includes a Leslie's cube, a thermopile, a galvanometer, and a low voltage power supply.
2. Fill the Leslie's cube with boiling water and place it in front of the thermopile.
3. Connect the thermopile to the galvanometer and the low voltage power supply.
4. Observe the deflection of the galvanometer needle and record the readings.
5. Repeat the experiment with different temperatures of water in the Leslie's cube.
6. Plot a graph of the galvanometer deflection against the temperature of the water in the Leslie's cube.
7. The graph should be a straight line, indicating that the total energy radiated is directly proportional to the fourth power of the temperature, thus verifying Stefan's law.

5. To determine resistance per unit length and specific resistance of a given resistance

- Resistance is a measure of the opposition to the flow of electric current through a conductor.
- The resistance of a conductor depends on its length, cross-sectional area, and the material it is made of.
- The resistance per unit length is the resistance of a conductor divided by its length.
- The specific resistance, also known as resistivity, is a property of the material the conductor is made of. It is defined as the resistance of a conductor of unit length and unit cross-sectional area.
- To determine the resistance per unit length and specific resistance of a given resistance, the following steps can be followed:
 1. Measure the length of the conductor using a ruler or measuring tape.
 2. Measure the cross-sectional area of the conductor using a micrometer screw gauge or vernier calipers.
 3. Connect the conductor in a circuit with a battery, an ammeter, and a voltmeter.
 4. Measure the current flowing through the conductor using the ammeter and the potential difference across the conductor using the voltmeter.
 5. Use Ohm's law to calculate the resistance of the conductor.
 6. Divide the resistance by the length of the conductor to obtain the resistance per unit length.
 7. Divide the resistance by the cross-sectional area of the conductor to obtain the specific resistance.
- It is important to note that the temperature of the conductor can affect its resistance. Therefore, it is important to conduct the experiment at a constant temperature.

Using Carey Foster's Bridge

Carey Foster's Bridge is a method used to measure the resistance of a wire or to compare two resistances. It is a modification of the Wheatstone bridge and is used when a high degree of accuracy is required.

Here are the steps to use Carey Foster's Bridge:

1. Connect the unknown resistance and the standard resistance in the two gaps of the bridge.
2. Connect a galvanometer between the junctions of the two resistances.
3. Connect a battery or other power source to the bridge.
4. Adjust the sliding contact on the wire until the galvanometer shows no deflection, indicating that the bridge is balanced.
5. Measure the length of the wire between the fixed and sliding contacts.
6. Calculate the resistance of the unknown wire using the formula: $R = (l * S) / L$, where R is the resistance of the unknown wire, l is the length of the wire between the fixed and sliding contacts, S is the resistance of the standard wire, and L is the total length of the wire.

By following these steps, you can accurately measure the resistance of a wire or compare two resistances using Carey Foster's Bridge.

6. To study the resonance condition of a series LCR circuit.

A series LCR circuit consists of an inductor (L), a capacitor (C), and a resistor (R) connected in series. The resonance condition of a series LCR circuit occurs when the inductive reactance (XL) is equal to the capacitive reactance (XC). At this point, the circuit is said to be in resonance and the impedance of the circuit is at its minimum value.

The resonance frequency (fr) of a series LCR circuit can be calculated using the formula:

$$f_r = 1 / (2 * \sqrt{LC})$$

At resonance, the voltage across the capacitor and the voltage across the inductor are equal in magnitude but opposite in phase. The current in the circuit is at its maximum value and is in phase with the voltage across the resistor.

The quality factor (Q) of a series LCR circuit is a measure of the sharpness of the resonance peak and is given by the formula:

$$Q = (2 * f_r * L) / R$$

The bandwidth (Δf) of a series LCR circuit is the range of frequencies over which the circuit exhibits resonance behavior and is given by the formula:

$$\Delta f = f_r / Q$$

In summary, the resonance condition of a series LCR circuit occurs when the inductive reactance is equal to the capacitive reactance. At resonance, the circuit

exhibits a minimum impedance, maximum current, and the voltage across the capacitor and inductor are equal in magnitude but opposite in phase. The resonance frequency, quality factor, and bandwidth of the circuit can be calculated using the formulas provided above.

7. To determine the electrochemical equivalent (ECE) of copper.

The electrochemical equivalent (ECE) of a substance is the mass of the substance deposited or dissolved at an electrode during electrolysis by the passage of one coulomb of electric charge. The ECE of copper can be determined by performing an experiment in which copper is deposited on an electrode during electrolysis.

The steps to determine the ECE of copper are as follows:

1. Set up an electrolytic cell with copper sulfate solution as the electrolyte and two copper electrodes.
2. Connect the electrodes to a power source and allow current to flow through the cell for a fixed amount of time.
3. Measure the mass of the copper deposited on the cathode.
4. Calculate the ECE of copper using the formula: $\text{ECE} = (\text{mass of copper deposited}) / (\text{charge passed})$.

The ECE of copper can be used to calculate the amount of copper that will be deposited on an electrode during electrolysis, given the amount of charge passed through the cell. This can be useful in various applications, such as electroplating and refining of copper.

8. To calibrate the given ammeter and voltmeter by potentiometer.

Calibration is the process of comparing the readings of an instrument with a known standard to determine the accuracy of the instrument. In this experiment, we will calibrate an ammeter and a voltmeter using a potentiometer.

The steps to calibrate the given ammeter and voltmeter by potentiometer are as follows:

1. Set up the circuit as shown in the diagram, with the ammeter, voltmeter, and potentiometer connected in series.
2. Adjust the potentiometer to obtain a known current through the circuit.
3. Record the reading on the ammeter and compare it with the known current.
4. If the ammeter reading is not equal to the known current, adjust the zero error of the ammeter accordingly.
5. Repeat the above steps for different values of current to obtain a calibration curve for the ammeter.
6. Similarly, adjust the potentiometer to obtain a known voltage across the voltmeter.

7. Record the reading on the voltmeter and compare it with the known voltage.
8. If the voltmeter reading is not equal to the known voltage, adjust the zero error of the voltmeter accordingly.
9. Repeat the above steps for different values of voltage to obtain a calibration curve for the voltmeter.

By following these steps, the given ammeter and voltmeter can be calibrated using a potentiometer. This will ensure that the readings obtained from these instruments are accurate and reliable.

9. To draw hysteresis (B-H curve) of a specimen in the form of a transformer and to

Hysteresis is the dependence of the state of a system on its history. In the context of magnetic materials, it refers to the lagging of the magnetization of a material behind the changes in the applied magnetic field. The hysteresis loop, also known as the B-H curve, is a graphical representation of this phenomenon.

To draw the hysteresis loop of a specimen in the form of a transformer, the following steps can be followed:

1. Set up the transformer with the specimen as the core, and connect it to an AC power source.
2. Connect an ammeter in series with the primary coil to measure the current, and a voltmeter across the secondary coil to measure the induced voltage.
3. Gradually increase the current in the primary coil and record the corresponding values of the induced voltage in the secondary coil.
4. Plot the induced voltage (B) against the current (H) to obtain the hysteresis loop.

The shape of the hysteresis loop depends on the magnetic properties of the specimen. Materials with a large hysteresis loop have a high coercivity and retentivity, meaning they require a large magnetic field to demagnetize and can retain their magnetization even after the applied field is removed. These materials are useful for making permanent magnets. On the other hand, materials with a small hysteresis loop have a low coercivity and retentivity, meaning they can be easily magnetized and demagnetized. These materials are useful for making transformer cores.

Hysteresis Loss

Hysteresis loss is the energy loss that occurs in a magnetic material due to the reversal of magnetization. It is a type of energy loss that occurs in ferromagnetic materials when they are subjected to a changing magnetic field. The energy loss is due to the movement of the magnetic domains within the material.

To determine the hysteresis loss, the following steps can be followed:

1. Measure the area of the hysteresis loop: The hysteresis loop is a graph that shows the relationship between the magnetic field strength and the magnetization of the material. The area of the hysteresis loop represents the energy loss per cycle.
2. Calculate the frequency of the magnetic field: The frequency of the magnetic field is the number of cycles per second. It can be calculated by measuring the time it takes for one complete cycle of the magnetic field.
3. Multiply the area of the hysteresis loop by the frequency of the magnetic field: The product of the area of the hysteresis loop and the frequency of the magnetic field gives the hysteresis loss per second.
4. Multiply the hysteresis loss per second by the time: To determine the total hysteresis loss, the hysteresis loss per second is multiplied by the time over which the magnetic field is applied.

In summary, to determine the hysteresis loss, the area of the hysteresis loop is measured, the frequency of the magnetic field is calculated, and the hysteresis loss per second is determined by multiplying the area of the hysteresis loop by the frequency of the magnetic field. The total hysteresis loss is then calculated by multiplying the hysteresis loss per second by the time over which the magnetic field is applied.

10. To measure high resistance by leakage method.

The leakage method is used to measure high resistance values. This method is based on the principle of measuring the leakage current flowing through the insulation resistance. The following are the steps involved in measuring high resistance by the leakage method:

1. The first step is to connect the high resistance to be measured in series with a low resistance of known value.
2. A voltage source is then applied across the series combination of the high and low resistances.
3. The voltage drop across the low resistance is measured using a voltmeter.
4. The leakage current flowing through the high resistance can be calculated using Ohm's law, by dividing the voltage drop across the low resistance by its resistance value.
5. The value of the high resistance can then be calculated by dividing the applied voltage by the calculated leakage current.

This method is commonly used to measure the insulation resistance of electrical equipment and cables. It is important to note that the accuracy of the measurement depends on the accuracy of the low resistance and the voltmeter used in the measurement. It is also important to ensure that the applied voltage is constant during the measurement.

Course outcomes:

- Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course.
- They are used to guide the design of the course, including the selection of course content, teaching methods, and assessments.
- Course outcomes should be specific, measurable, and achievable within the scope of the course.
- They should be aligned with the overall goals of the program and the institution.
- Course outcomes can be used to evaluate the effectiveness of the course and to identify areas for improvement.
- They can also be used to communicate the value of the course to students, employers, and other stakeholders.
- Course outcomes should be reviewed and updated regularly to ensure that they remain relevant and aligned with the needs of students and the broader community.

At the end of the course, students will be able to:

- Demonstrate knowledge and understanding of the subject matter.
- Apply the concepts and theories learned in the course to real-world situations.
- Analyze and evaluate information critically and effectively.
- Communicate their ideas and arguments clearly and effectively.
- Work collaboratively with others to achieve common goals.
- Demonstrate ethical and professional behavior.
- Develop skills for lifelong learning and personal growth.

CO CO Statement Bloom's

CO (Course Outcome) and CO statement are used to describe the expected outcomes of a course. These outcomes are usually defined in terms of knowledge, skills, and attitudes that students are expected to have acquired by the end of the course.

Bloom's taxonomy is a framework for categorizing educational goals and objectives. It is often used to help design assessments, curriculum, and instructional methods. The taxonomy consists of six levels of cognitive learning, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills. These levels are:

1. Remembering: recalling information from memory.
2. Understanding: comprehending the meaning of information.
3. Applying: using information in a new situation.
4. Analyzing: breaking down information into parts and examining their relationships.
5. Evaluating: making judgments about the value of information.

6. Creating: combining information to form a new whole.

CO statements can be written using Bloom's taxonomy to ensure that the course outcomes are aligned with the desired level of cognitive learning. For example, a CO statement for a course on programming might be: "By the end of this course, students will be able to apply their knowledge of programming concepts to design and implement a simple program (Applying)." This CO statement specifies that the desired outcome of the course is for students to be able to apply their knowledge of programming concepts, which corresponds to the third level of Bloom's taxonomy.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels are commonly used in construction, carpentry, and surveying.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness.
- Laser levels project a level line using a laser beam.
- Water levels use the principle that water will always find its own level to determine levelness.
- Levels can vary in size, from small pocket-sized levels to large, heavy-duty levels used in construction.
- It is important to use a level when installing floors, hanging pictures, or building structures to ensure that they are level and plumb.

CO-1 Apply the principle of interference and diffraction to find the wavelength of

Interference and diffraction are two phenomena that can be used to determine the wavelength of light. Here are the steps to apply these principles to find the wavelength of light:

1. **Interference:** Interference occurs when two or more waves overlap and combine to form a new wave. The principle of interference can be used to find the wavelength of light by using a device called an interferometer. In an interferometer, a beam of light is split into two beams that travel different paths and then recombine. The difference in the path lengths of the two beams causes the waves to interfere with each other, producing an interference pattern. By measuring the interference pattern, the wavelength of the light can be determined.
2. **Diffraction:** Diffraction is the bending of waves around obstacles or through openings. The principle of diffraction can be used to find the wavelength of light by shining a beam of light through a narrow slit or a diffraction grating. The light waves spread out as they pass through

the slit or grating, producing a diffraction pattern. By measuring the diffraction pattern, the wavelength of the light can be determined.

In summary, the principles of interference and diffraction can be applied to find the wavelength of light by using an interferometer or a diffraction grating and measuring the resulting interference or diffraction pattern.

Monochromatic and Polychromatic Light

- **Monochromatic light** refers to light that consists of a single wavelength or color. This type of light is produced by lasers, LEDs, and certain types of lamps.
- **Polychromatic light**, on the other hand, consists of multiple wavelengths or colors. This type of light is produced by most light sources, including the sun, incandescent bulbs, and fluorescent lamps.
- Monochromatic light is often used in scientific and medical applications, such as spectroscopy and microscopy, where precise control over the wavelength of light is required.
- Polychromatic light is used in everyday applications, such as lighting and displays, where a broad spectrum of colors is desired.
- The difference between monochromatic and polychromatic light can also be observed in the way they interact with materials. Monochromatic light will produce sharp, well-defined diffraction patterns, while polychromatic light will produce more complex, overlapping patterns.
- In summary, monochromatic light consists of a single wavelength, while polychromatic light consists of multiple wavelengths. Each type of light has its own unique properties and applications.

CO-2 Compute and analyze various electrical and electronic properties of a given

1. To compute and analyze various electrical and electronic properties of a given, one must first understand the basic concepts of electricity and electronics.
2. Electrical properties refer to the characteristics of a material or system that determine its behavior in the presence of an electric field. These properties include resistance, capacitance, and inductance.
3. Electronic properties refer to the characteristics of a material or system that determine its behavior in the presence of an electric current. These properties include conductivity, mobility, and carrier concentration.
4. To compute these properties, one must use various mathematical equations and formulas. For example, Ohm's law can be used to compute the resistance of a material, while the capacitance of a capacitor can be computed using the formula $C = Q/V$.
5. To analyze these properties, one must use various tools and techniques such as oscilloscopes, multimeters, and spectrum analyzers. These tools

allow one to measure and visualize the electrical and electronic properties of a given material or system.

6. By computing and analyzing these properties, one can gain a deeper understanding of the behavior of a given material or system in the presence of electric fields and currents. This knowledge can be used to design and optimize electronic devices and systems.

Material Analysis by Using Various Experiments

Material analysis is the process of determining the chemical composition, structure, and properties of a material. This information is used to understand the behavior of the material under different conditions and to develop new materials with improved properties. There are several experiments that can be used to analyze materials, including:

1. **X-ray diffraction (XRD):** This technique is used to determine the crystal structure of a material. It involves shining X-rays onto a sample and measuring the diffraction pattern produced. The pattern provides information about the arrangement of atoms in the material.
2. **Scanning electron microscopy (SEM):** This technique is used to study the surface of a material. It involves scanning a focused beam of electrons across the surface of the sample and measuring the electrons that are scattered. The resulting image provides information about the topography and composition of the surface.
3. **Transmission electron microscopy (TEM):** This technique is used to study the internal structure of a material. It involves passing a beam of electrons through a thin sample and measuring the electrons that are transmitted. The resulting image provides information about the internal structure of the material.
4. **Fourier transform infrared spectroscopy (FTIR):** This technique is used to study the chemical composition of a material. It involves shining infrared light onto a sample and measuring the absorption of the light. The absorption spectrum provides information about the chemical bonds present in the material.
5. **Thermogravimetric analysis (TGA):** This technique is used to study the thermal stability of a material. It involves heating a sample and measuring the change in its mass as a function of temperature. The resulting data provides information about the decomposition of the material.

These are just a few of the many experiments that can be used to analyze materials. The choice of experiment depends on the specific properties of the material that need to be studied.

CO-3 Verify different established laws with the help of optical and electrical

Optical and electrical experiments can be used to verify different established laws in physics. Some of these laws include:

1. **Ohm's Law:** This law states that the current flowing through a conductor is directly proportional to the voltage applied across it, provided the temperature remains constant. This law can be verified using a simple circuit consisting of a battery, a resistor, an ammeter, and a voltmeter.
2. **Snell's Law:** This law describes the relationship between the angles of incidence and refraction when light passes through the interface between two different media. This law can be verified using a simple experiment involving a light source, a transparent block of material with known refractive index, and a protractor.
3. **Kirchhoff's Laws:** These laws describe the conservation of charge and energy in electrical circuits. Kirchhoff's Current Law (KCL) states that the algebraic sum of currents entering a node is zero, while Kirchhoff's Voltage Law (KVL) states that the algebraic sum of voltages around a closed loop is zero. These laws can be verified using a simple circuit consisting of a battery, resistors, and a multimeter.
4. **The Law of Reflection:** This law states that the angle of incidence is equal to the angle of reflection when light is reflected off a smooth surface. This law can be verified using a simple experiment involving a light source, a flat mirror, and a protractor.

These are just a few examples of the many established laws that can be verified using optical and electrical experiments. By conducting these experiments, students can gain a deeper understanding of the fundamental principles of physics.

Experiments

An experiment is a scientific procedure undertaken to make a discovery, test a hypothesis, or demonstrate a known fact. Here are some key points to consider when conducting an experiment:

1. **Hypothesis:** A hypothesis is a proposed explanation for a phenomenon. It is important to have a clear and testable hypothesis before conducting an experiment.
2. **Variables:** Variables are factors that can be changed or controlled in an experiment. It is important to identify and control variables to ensure the accuracy of the results.
3. **Control group:** A control group is a group in an experiment that does not receive the treatment being tested. It is used as a baseline to compare the results of the treatment group.
4. **Data collection:** Data collection is the process of gathering and measuring information. It is important to have a clear and consistent method for collecting data to ensure the accuracy of the results.

5. **Analysis:** Analysis is the process of examining and interpreting the data collected in an experiment. It is important to use appropriate statistical methods to analyze the data and draw conclusions.
6. **Replication:** Replication is the process of repeating an experiment to confirm the results. It is important to replicate experiments to ensure the reliability of the results.

Apply

1. The term “apply” can refer to the act of putting something into operation or use, such as applying a rule or principle.
2. It can also refer to the act of making a formal request or submission, such as applying for a job or a loan.
3. In the context of mathematics or computing, “apply” can refer to the act of using a function or operation to a set of values or arguments.
4. In the context of cosmetics or medicine, “apply” can refer to the act of putting a substance onto a surface, such as applying makeup or a topical medication.
5. In general, the term “apply” implies a deliberate and purposeful action, often with the goal of achieving a specific result or effect.

CO-4 Determine and calculate various physical properties of a given material by using

1. **Density:** The density of a material is defined as its mass per unit volume. It can be calculated by dividing the mass of the material by its volume.
2. **Elasticity:** Elasticity is the ability of a material to return to its original shape after being deformed. The modulus of elasticity, also known as Young’s modulus, is a measure of the stiffness of a material and can be calculated by dividing the stress by the strain.
3. **Thermal conductivity:** Thermal conductivity is a measure of a material’s ability to conduct heat. It can be calculated by dividing the heat flux by the temperature gradient.
4. **Electrical conductivity:** Electrical conductivity is a measure of a material’s ability to conduct electricity. It can be calculated by dividing the current density by the electric field.
5. **Hardness:** Hardness is a measure of a material’s resistance to deformation or indentation. It can be measured using various methods, such as the Brinell, Rockwell, or Vickers hardness tests.
6. **Melting point:** The melting point of a material is the temperature at which it changes from a solid to a liquid state. It can be determined experimentally by heating the material and observing the temperature at which it melts.
7. **Tensile strength:** Tensile strength is a measure of the maximum stress a material can withstand before breaking when subjected to a tensile force. It can be calculated by dividing the maximum load by the cross-sectional

area of the material.

Various Experiments

There are many different types of experiments that can be conducted in various fields of study. Some common types of experiments include:

1. **Controlled experiments:** These experiments involve manipulating one or more independent variables to observe the effect on a dependent variable. The goal is to establish a cause-and-effect relationship between the independent and dependent variables.
2. **Natural experiments:** These experiments take advantage of naturally occurring events or situations to study the effect of an independent variable on a dependent variable. The goal is to observe the effect of the independent variable in a real-world setting.
3. **Field experiments:** These experiments are conducted in a natural setting, outside of a laboratory. The goal is to study the effect of an independent variable on a dependent variable in a real-world setting.
4. **Quasi-experiments:** These experiments involve comparing groups that are not randomly assigned. The goal is to study the effect of an independent variable on a dependent variable, while controlling for other factors that may influence the results.
5. **Longitudinal experiments:** These experiments involve collecting data from the same group of participants over an extended period of time. The goal is to study changes in the dependent variable over time.
6. **Cross-sectional experiments:** These experiments involve collecting data from different groups of participants at the same point in time. The goal is to compare the groups on the dependent variable.

These are just a few examples of the many different types of experiments that can be conducted. The specific type of experiment chosen will depend on the research question, the variables being studied, and the resources available to the researcher.

Apply

1. The term “apply” can refer to the act of putting something into operation or use, such as applying a rule or a principle.
2. It can also refer to the act of making a formal request or submission, such as applying for a job or a loan.
3. In the context of mathematics, “apply” can refer to the act of using a mathematical operation or function to a given value or set of values.
4. In the context of computer science, “apply” can refer to the act of calling a function with a given set of arguments.

5. In general, “apply” refers to the act of putting something into action or use, or making a request or submission.

CO-5 Study and estimate the performance and parameter of given equipment by using

1. To study and estimate the performance and parameter of given equipment, it is important to first understand the specifications and capabilities of the equipment.
2. This can be done by referring to the equipment’s manual or technical documentation, which should provide detailed information on its performance and parameters.
3. Once the specifications are understood, the equipment can be tested under various conditions to measure its performance and determine its parameters.
4. This can be done by conducting experiments or simulations, and collecting data on the equipment’s performance.
5. The data collected can then be analyzed to estimate the equipment’s performance and parameters.
6. It is important to conduct these tests and analyses in a controlled and systematic manner to ensure accurate and reliable results.
7. The results of the study can be used to optimize the equipment’s performance, and to make informed decisions on its use and maintenance.

Graphical and Computational Analysis

Graphical and computational analysis are two methods used to analyze data and solve problems. Here are some key points to consider:

1. **Graphical analysis** involves the use of graphs and charts to represent data visually. This can help identify trends, patterns, and relationships between variables.
2. **Computational analysis** involves the use of mathematical algorithms and computer programs to analyze data and solve problems. This can be useful for handling large datasets and complex calculations.
3. Both methods can be used together to provide a more comprehensive analysis. For example, computational analysis can be used to generate graphs and charts for graphical analysis.
4. Graphical analysis can be useful for presenting data in a way that is easy to understand, while computational analysis can provide more detailed and accurate results.
5. The choice of method will depend on the specific problem being addressed and the type of data being analyzed. It is important to carefully consider the strengths and limitations of each method before deciding which one to use.

Apply

- The term “apply” can have several meanings depending on the context in which it is used.
- It can mean to make a formal request, usually in writing, for something such as a job, admission to a school, or a loan.
- It can also mean to put or spread something, such as a substance or a force, onto or over something else.
- In mathematics, it can mean to use a rule or formula to calculate or solve a problem.
- In computer science, it can refer to the process of using a function or operation to a set of data or arguments.
- In general, the term “apply” refers to the act of putting something into action or practice.

Reference Books

- Reference books are a type of non-fiction book that provides information on a specific subject.
- They are often used for research and study purposes.
- Examples of reference books include dictionaries, encyclopedias, almanacs, and atlases.
- Reference books are usually organized in a way that makes it easy to find the information you need.
- They often contain an index or table of contents to help you locate specific information.
- Reference books are usually not meant to be read cover-to-cover, but rather to be consulted for specific information.
- They are often updated regularly to ensure that the information they contain is current and accurate.
- Reference books can be found in libraries, bookstores, and online.
- They are an important resource for students, researchers, and anyone seeking to learn more about a particular subject.

1. Practical Physics- K. K. Dey & B. N. Dutta (Kalyani Publishers New Delhi)

- “Practical Physics” is a book written by K. K. Dey and B. N. Dutta.
- It is published by Kalyani Publishers, New Delhi.
- The book is a comprehensive guide to practical physics experiments and measurements.
- It covers a wide range of topics, including mechanics, heat, optics, electricity, and magnetism.
- The book is written in a clear and concise manner, with numerous diagrams and illustrations to aid understanding.
- It is a valuable resource for students studying physics at the undergraduate level.

- The book provides detailed instructions for performing experiments and taking measurements, as well as guidance on how to analyze and interpret data.
- Overall, “Practical Physics” by K. K. Dey and B. N. Dutta is a highly recommended book for anyone interested in gaining a deeper understanding of the practical aspects of physics.

2. Engineering Physics-Theory and Practical- Katiyar & Pandey (Wiley India)

- Engineering Physics-Theory and Practical is a book written by Katiyar & Pandey and published by Wiley India.
- The book covers the fundamental concepts of physics and their applications in engineering.
- It is designed to provide a comprehensive understanding of the subject to engineering students.
- The book is divided into two parts: Theory and Practical.
- The Theory part covers topics such as mechanics, thermodynamics, electromagnetism, optics, and modern physics.
- The Practical part includes experiments and practical applications of the concepts covered in the Theory part.
- The book is written in a clear and concise manner, making it easy for students to understand and learn from.
- It is a valuable resource for engineering students preparing for exams or seeking to deepen their understanding of physics and its applications in engineering.

3. Engineering Physics Practical- S K Gupta (Krishna Prakashan Meerut)

- Engineering Physics Practical is a book written by S K Gupta and published by Krishna Prakashan Meerut.
- The book is intended for students studying engineering and covers practical aspects of physics.
- It includes experiments and practical applications of physics concepts relevant to engineering.
- The book is designed to help students develop practical skills and gain hands-on experience in applying physics to engineering problems.
- It is a valuable resource for students preparing for exams and for those who want to deepen their understanding of physics in an engineering context.